Muhammad Talha Atif

Backend Engineer | Node.js · Express · NestJS

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SUMMARY

Backend engineer with 1+ years of production experience building scalable server-side systems using Node.js, Express, and NestJS. Designed and optimized RESTful and event-driven APIs, relational and NoSQL databases, and cloud-native deployments. Delivered a NestJS-powered AI booking platform with async message queues and real-time processing, and a full-stack social platform with an Express backend and Next.js frontend. Comfortable owning backend services end-to-end from database schema to deployment, with a strong focus on performance, clean architecture, and reliability. Also proficient in Next.js frontend development for seamless API integration.

WORK EXPERIENCE

Full Stack Engineer *Dec '25 – Present*
Intimec via MicroTechX Remote

Engineered the Next.js dashboard for an enterprise Identity Governance platform, implementing the frontend design system, reusable component library, and secure API integration.

- Standardised the frontend theme, color system, and shared component library across the Next.js dashboard, eliminating UI drift and establishing team-wide conventions.
- Improved client-side performance by reducing bundle sizes and implementing efficient data-fetching strategies with server-side rendering and static generation.
- Collaborated with backend engineers to define REST API schemas for access requests, audit logging, and entitlement management, ensuring clean frontend-backend contracts.
- Integrated role-based access control (RBAC) and JWT-based authentication flows into the UI, enhancing platform security and user experience.

Full Stack Engineer *Sep '25 – Jan '26*
Digital Upgraders LLC Remote

Built and maintained backend services and frontend features for a production social platform serving photographers, videographers, and creative professionals.

- Developed and optimized RESTful APIs with Node.js and Express for feeds, notifications, authentication, and media workflows, handling thousands of daily active users.
- Cut feed load time from 3s to 700ms by restructuring database queries, adding MongoDB indexes, minimizing payload sizes, and implementing Redis caching.
- Fixed duplicate notifications by redesigning concurrent write flows with proper MongoDB transactions, restoring data consistency across the system.
- Led a zero-downtime infrastructure migration from AWS to Fly.io, reworking Docker configurations, CI/CD pipelines, and environment variables on a live production platform.

PROJECTS

Moodly | AI Experience Booking Platform (NestJS, TypeScript, RabbitMQ) | Demo

- Architected a modular NestJS backend with Guards, Interceptors, and Pipes for authentication, validation, and logging across 15+ RESTful endpoints.
- Designed a multimodal emotion detection pipeline that processes voice and facial images through separate ML models, routing results via RabbitMQ to a booking recommendation engine.
- Implemented JWT-based authentication with refresh tokens and role-based access control (RBAC) for hosts and users, achieving 90%+ test coverage on core business logic.
- Built a React Native cross-platform frontend with one-tap booking flows, real-time mood visualisations, and deep backend integration via structured API responses.

CunoCovo | Social Platform (Express, Node.js, MongoDB, Next.js) | Demo

- Developed a RESTful API with Express.js, including rate limiting, error handling middleware, and JWT authentication, documented with Swagger for team and client use.
- Designed MongoDB schemas for user profiles, posts, and interactions, adding compound indexes that improved query performance by 60%.
- Built the Next.js frontend with server-side rendering, dynamic routes, and API route handlers, reducing time-to-interactive and improving SEO.
- Integrated real-time notification delivery using Socket.io, synchronised with the Express backend to provide instant updates across the platform.

SKILLS

Languages TypeScript, JavaScript (ES6+), Python

Backend Node.js, Express.js, NestJS, REST APIs, gRPC, WebSockets, RabbitMQ

Frontend Next.js, React.js, React Native, Tailwind CSS, Shadcn UI, Framer Motion

Architecture Microservices, Event-Driven Architecture, Distributed Systems, Kafka (basics)

Databases MongoDB, PostgreSQL, Redis, MySQL, Supabase, Firebase

Security JWT, OAuth 2.0, RBAC, Rate Limiting, Input Validation, Helmet

Cloud AWS (S3, EC2, RDS), Fly.io, Vercel, Dokploy

DevOps Docker, GitHub Actions, CI/CD, Git

Tools Postman, Swagger, VS Code, MongoDB Compass

AWARDS

Runner-up, App Development Hackathon (2025) *May '25*
Organized by Punjab University

Ranked, ICPC Asia Topi Online Programming Contest (2024) *Nov '24*
Organized by International Collegiate Programming Contests

EDUCATION

Bachelor's degree in Computer Science, University of Management and Technology – UMT *Oct '21 – Aug '25*